Gestalt, Harry Potter's spells and a bathtub for Jennifer Lopez

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Abstract

"It's time for the United States to consider establishing higher standards for math teachers if the nation is going to break its "vicious cycle" of mediocrity[1] ..." "Michigan State University is committed to giving students the education they need to prepare them to be citizens in our highly technological and scientific world[2]." These two samples reveal the direction of the Zeitgeist in the current educational science environment. The main goal of this article is to show the urgent need of better understanding of core science ideas and practices.

[1]W.Schmidt, Want better math teachers? Train them better, scholar argues, 2011, MSU News, http://research.msu.edu/tags/william-schmidt. He is a member of the National Academy of Education, a fellow of the American Educational Research Association, Director of MSU PROM/SE, EPC at MSU.

[2]J.Krajcik, A speech about CREATE in online forum, http://create4stem.msu.edu/story/institute -director-speaks-about-create-online-forum. He served as a president of National Association for Research in Science Teaching, a fellow of American Association for the Advancement of Science, Director of the MSU Institute CREATE for STEM.

Aims and Targets

Once, one day of March of 2013 I was participating in a Woodrow Wilson Teaching Competition passing it up from the narrow valley of rules for applicants into which it opens, by following the prescribed directions of making a presentation and watching a specially designated 'teaching art video' on Utube for being cognizant of advanced math class inventory. How these two components of the preparation, straight and simple enough as they were, had been assembled in results of the preparation process with having been bringing a certain realizing of ideas about math education methodology point (with multi facets still almost inaccessible) is a problem in the solution of which there would be greater pedagogical satisfaction than scientific advantage.

The mentioned video was dedicated to a virtuoso teaching in a real school class mathematics. I detected myself staring more solicitously at the certain class event than I could find warrant for doing. I was unable to repress a certain feeling of apprehension that I saw the rationale of my apprised incursion in the subject with my devotion to the teacher's duty understood.

Mentioned advanced mathematical class had a deal with mixes with percentages in an usual problem discernible consideration in it's classic form - what percentage for a new mix we obtain "babeling" amount N of A percent salt solution with amount M of B percent (the same salt) solution. The nice practical illustration was provided before the class obtained a receipt how to execute the task. Without any abatement of my affectionate solicitude for the good teacher's action I began to regard the results as at the edge of perfunctory act. The mentioned receipt was based on proportionality rule (and the teacher didn't use a diaphanous opportunity to explain the elementary mechanics of the equilibrium equation (a yoke's balance[3]) was prompted as an easy and effective trick (without rigorous mathematical inference). It actually militates against real understanding of mixes. Through this receipt-trick an encouraged class got a skill which superseded real understanding of calculation of mixes. The evident premise of this conclusion is in shifting aside from the standard conditions of the problem - zealots of the skill obtain pretty predicaments if consider mixes of interacting stuff or mixes of different salts and many other natural cases. That will take away the skill having falling into desuetude.

The title chosen for my presentation could be "Think in and out of the box" or even "Think in and out of the garbage box" to reveal an important link with gestalt [1]. The whole (pupil's current math ability) as it is has a discrepancy from the sum of its parts (like if enhancing skills were juxtaposed in prescribed order). The phenomena when quantity becomes quality. Among interesting examples of this phenomena is how Napoleon compares the combat between the French cavalry and the Mamelukes: "Two Mamelukes were undoubtedly more than a match for three Frenchmen; 100 Mamelukes were equal to 100 Frenchmen; and 1,000 Frenchmen invariably defeated 1,500 Mamelukes." [2].

The pupil's math ability accrues through "skills and knowledge" coalescing process, amalgamating skills and knowledge not just by aggrandizing by numerous skills. The skill within whole math ability differs from the skill on its own. Synthetic analysis must be in charge. Among popular literature closer to the subject considered I can pick up the following: when doctor House was young he was an aristocratic drone Wooster [3] having his personal valet Jeeves and both of them were cognizant of the dictum, "practically incessant merriment", that "education is a drawing out, not a putting in" (this is about the gestalt's "shape" figuring out).

Next I chose that a protracted voyage through a sea of mathematics educational pedagogy

[3]Unlike "the yoke of enmity": Matthew 11 v 28.

theories (crossing new and obliterated ones) would not be agreeable and beneficial in the following[4]. I intend to turn to the genius of human thinking. I mean the greatest theoretical physicist, who won the Nobel Prize in Physics in first quarter of XX century and is famous as the founder of XX century physics, and who contributed considerably to extend the theory of relativity (special), the relativistic thermodynamics, optics, quantum theory. I imagine that the reader has already recognized the name - Max Plank. I am using one of his speeches [4] where he demonstrated bright ideas for mathematical education. He said that one good understanding of a mathematical sentence is better than to remember (and to know how to apply) ten formulas. Though the knowledge without skill doesn't have a value but the theory should never be substituted by the set of skills because it will be helpless when the new unusual condition or situation comes.

If you feel as my statements upset the applecart the following will dispel your apprehension.

A trick (spell) as a skill

Here we consider the *point d'appui* of the case (The classical approach considered can be found in many text-books, for instance, in [5]).

For two non-negatives X_1, X_2 from an evident inequality $\left(\sqrt{X_1}-\sqrt{X_2}\right)^2 \geq 0$ it is easy to obtain the inequality for the geometric mean (lest side of the inequality) and for the arithmetic mean (right side):

$$\sqrt[2]{X_1 X_2} \leq \frac{X_1+X_2}{2},$$

it is also shows that these means are equal if and only if all X's are equal.

If we unleash our mathematical intuition we can expand this inequality for three (non-negative) variables:

$$\sqrt[3]{X_1 X_2 X_3} \leq \frac{X_1+X_2+X_3}{3}$$

or

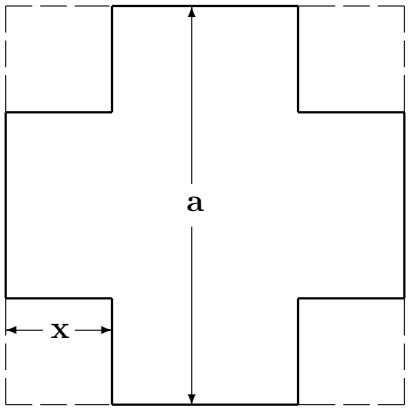
$$X_1 X_2 X_3 \leq \frac{(X_1+X_2+X_3)^3}{27} \tag{1}$$

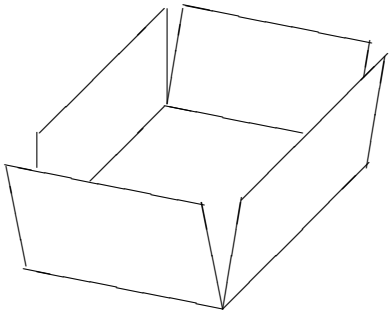
As before the equality can be obtained if and only if all X's are equal.

We don't have a target to prove inequality (1) right now (in [6] you can find 12 ways to prove it in general case). We have another target - to apply to the following engineering problem:

There is a square (metallic) sheet of side a. We want to cut out four little corner squares of side x in order to obtain a box. We want to obtain the maximal volume of the box.

[4]For instance, one of the closest considerations was in [8]. Stable URL: http://www.jstor.org/stable/40539326





It is natural to see that if we cut too little then the box will be very shallow, if cut too much - the box will be like a pipe (with one end closed).

We expect to have an optimum when cutting properly.

The volume of the box is $V = x(a-2x)^2$. Let us make a little trick. We consider $4V$ - four times volume:

$$4V = 4x(a-2x)^2 = \underbrace{4x}_{X_1} \times \underbrace{(a-2x)}_{X_2} \times \underbrace{(a-2x)}_{X_3}$$

according to (1) it is less or equal to

$$\leq \frac{(X_1+X_2+X_3)^3}{27} = \frac{(4x+(a-2x)+(a-2x))^3}{27} = \frac{(2a)^3}{27},$$

i.e., $4V$ is less or equal to the constant $\frac{(2a)^3}{27}$. We can reach this maximal volume's value if all X's are equal. It means that $a-2x = 4x$, or $x = a/6$. So, that's it. We are done with it. We obtained the result we needed. And our pupils obtained a skill how to maximize the box volume, and now they know that the maximal volume is $\frac{(2a)^3}{27\cdot 4}$.

Mr.Wooster noted: "Get off the mark, and your public is at a loss" [3]. Indeed, our smart talking, nice and kind smiling and light calculating can inculcate our pupils that they are in, in the case. As if they understand the case. Actually they have obtained only something like Harry Potter's spell against the case - without getting how the magic works.

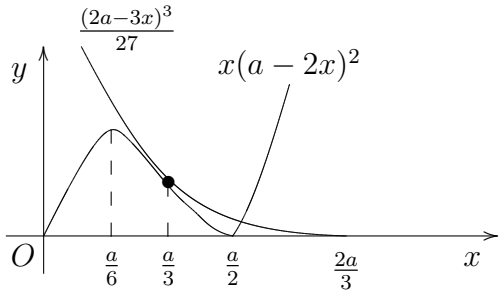
Let the pupil to do that himself. He, happy or unhappy in the opportunity of exploration and adventure, probably, will take the straight way using (1) and working with the box volume and will estimate the volume with the expression $(2a-3x)^3/27$:

$$V = x(a-2x)^2 \leq \frac{(x+(a-2x)+(a-2x))^3}{27} = \frac{(2a-3x)^3}{27}. \qquad (2)$$

That's right but it depends on x. A way that bore no resemblance to the one by which he had made first proof. It may be assumed - whatever the value of the assumption in connection with that is said to have occurred - that his mind was occupied with reflections of his understanding the receipt given that had been written in his mind.

Shocking in this denouement if he thinks in the box he can say: I can reach the upper bound making all X's equal. The apposite matter of the previous proof was not brought before him. It means that $a-2x = x$ or $x = a/3$. Here he gets a pretty predicament.

Of course, the matter is that the two cubic curves: the volume function and the upper function (the right hand side of (2)) meet each other only when $x = a/3$. For other x's the upper function in unreachable (for the real values of x: $0 \leq 2x \leq a$):



The skill obtained didn't transfigure in a knowledge. Gestalt is not realized.

A bathtub for Jennifer Lopez

Let us apply the approach considered to the case of a rectangular of size $a \times b$. We want to cut out four little corner squares of side x in order to obtain kind of a bathtub after folding. We want to obtain the maximal volume for the box. It is easy to see that if we cut too little the bathtub depth (up to the bee's knees) cannot be satisfactory (neither if we cut too much) giving us the heeby jeebys. Getting the maximal volume of the bathtub we suggest that it not for using by an anorexic high-fashion top-model but by a Hollywood star - naturally curvy with a healthy winning shape - like Jennifer Lopez.

The volume of the bathtub is $V = x(a-2x)(b-2x)$:

$$V = x(a-2x)(b-2x) = \underbrace{x}_{X_1} \times \underbrace{(a-2x)}_{X_2} \times \underbrace{(b-2x)}_{X_3}.$$

We can realize that in the case of a square we used a managing parameter z considering the $z \times V$ and putting the proper value of this parameter as $z = 4$. Such presumption transcends the mere effrontery limits of the problem considered and passes into the more wide bound area. Namely, we consider two positive parameters: z and y in the expression

$$zyV = x \times z(a-2x) \times y(b-2x) = \underbrace{x}_{X_1} \times \underbrace{z(a-2x)}_{X_2} \times \underbrace{y(b-2x)}_{X_3}$$

trying to find values of x, y, z to maximize the volume V of the bathtub. According to (1) this means that the sum of these X's must be constant (referring to x) and these X's must be equal:

$$\begin{cases} -2xz - 2xy + x = 0 \\ z(a-2x) = y(b-2x) = x \end{cases}$$

So, we have after light calculations:

$$\begin{cases} z = \frac{1-2y}{2} \\ z = \frac{x}{a-2x} \\ y = \frac{x}{b-2x} \end{cases}$$

It gives us the quadratic equation in terms of x: $12x^2 - 4(a+b)x + ab = 0$ with roots

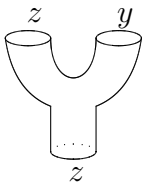
$$\frac{4(a+b) \pm \sqrt{16(a+b)^2 - 48ab}}{24}. \tag{3}$$

The determinant (the expression under the radical) is positive - it equals to $16(a^2 - ab + b^2)$.

Let us suppose that $a < b$. Then a light calculating gives that $a - 2 \cdot \frac{4(a+b)+\sqrt{16(a^2-ab+b^2)}}{24} < 0$, i.e. this root is too big. It is easy to see that the $\mathbf{x} = \frac{4(a+b)-\sqrt{16(a^2-ab+b^2)}}{24}$ is positive and less than $a/2$: $0 < \mathbf{x} < a/2$, i.e., $\mathbf{x}$ is a solution of the problem considered. More, if $a = b$ then we have obtained solution for the square.

Conclusion

Only after the pupil realized the parametric nature of the z in the square case he will be able to extend the used approach to the two-parametric model (and then - to several parameters or even infinity-dimensional model). This "parametric gestalt" we can draw like



meaning that one-parametric case is a base for the two-parametric case (with new degrees of freedom). The quantity (two parameters) brings a new quality (allowing to solve a new problem). The skill, based on the trick, expanded to be the new knowledge, new math charms and graces sees our pupil today. After "parametric augmentation" gets applied to the problem the desired solution becomes a set of merely ordinal calculation (regarding as well to the mentioned maximal volume of the tub). New understanding has been settled with all possible differences of opinion thereunto appertaining.

Why just knowing how to do tricks in Harry Potter fashion without knowing the matter of magic is a bad option? I want to provide an extreme (in some sense) example. The Moscow State University has a beautiful and very tall building resembling a rocket with student dormitories (high, multistoried) as wings of the rocket. About three decades ago when I was a post-graduate student at Moscow State University (mechanical-mathematical department) and lived in the math student dormitory (the top of the wing) I (as others) was informed about real suicidal acts of students (mostly mathematician students) who jumped from the window seeking death. As we could understand the one main issue was the extreme difficulty of mathematics (together with a tragedy in personal life and so on). I made a little investigation and realized that the suicidal student cases with hard studying issues were much more often in engineering institutes. Now, let us take in account, that mathematics courses for engineering students are mostly pragmatic, without time for real understanding of the mathematical matter, just a set of "magical" practical skills. That hurts the personal psyche.

I want to repeat Max Planck's idea that a good understanding of a mathematical sentence is better than just remembering how to apply formulas. Also, in [8] we can find the closer idea - for undergraduate students to learn a generic instantiation of a mathematical concept is more promising than to learn one or multiple concrete instantiations of the same concept.

I rest my case. For the sake of bringing more attention to this case, the intensive "Ambrose-Biercing" ([7]) of language in the article was used.

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